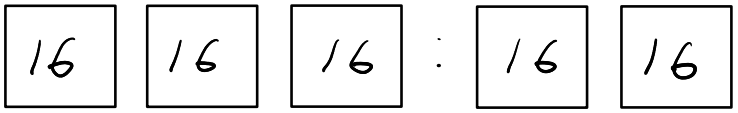
1 Will and Olly share £80 in the ratio 3 : 2

Work out how much each of them get.



$80 \div 5 = 16$

$3 \times 16 = 48$

$2 \times 16 = 32$

Will £ 48

Olly £ 32

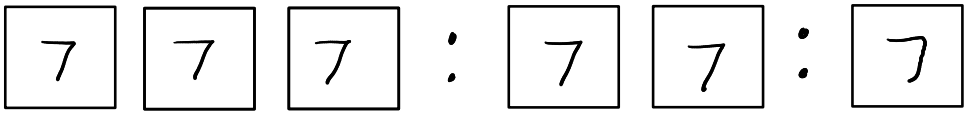
(Total for Question 1 is 3 marks)

---

2 Molly, Paige and Demi share 42 sweets in the ratio 3 : 2 : 1

Work out the number of sweets that each of them receives.

Molly Paige Demi



$42 \div 6 = 7$

Molly 21 sweets

Paige 14 sweets

Demi 7 sweets

(Total for Question 2 is 3 marks)

---

3 *ABC* is a straight line.



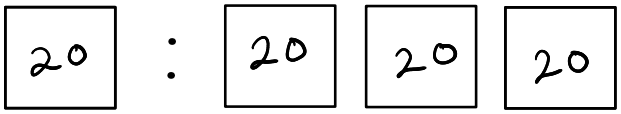
The length of *BC* is three times the length of *AB*.

AC = 80 metres.
Work out the length *BC*.

AB : BC

1 : 3

$80 \div 4 = 20$



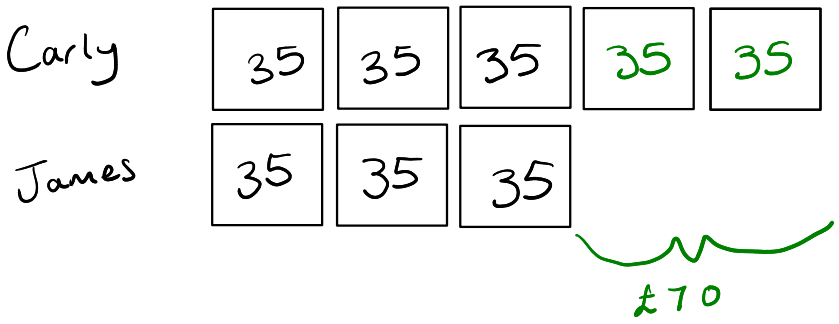
20 : 60

60 metres

(Total for Question 3 is 3 marks)

4 Carly and James share some money in the ratio 5 : 3
Carly gets £70 more than James.

Work out how much money James gets.



70 ÷ 2 = 35

3 x 35 = 105

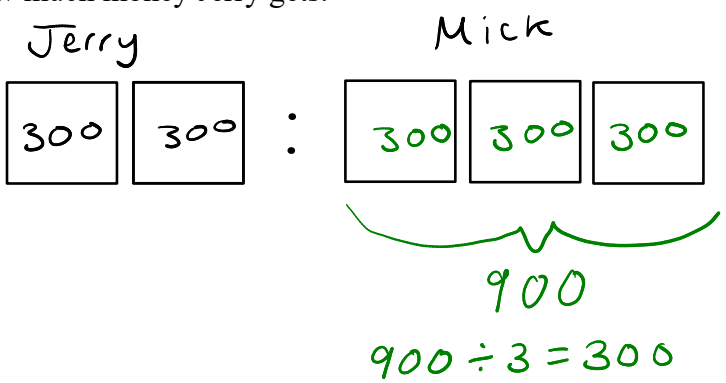
£ 105

(Total for Question 4 is 3 marks)

---

5 Jerry and Mick share some money in the ratio 2 : 3
Mick gets £900

Work out how much money Jerry gets.



900 ÷ 3 = 300

2 x 300 = 600

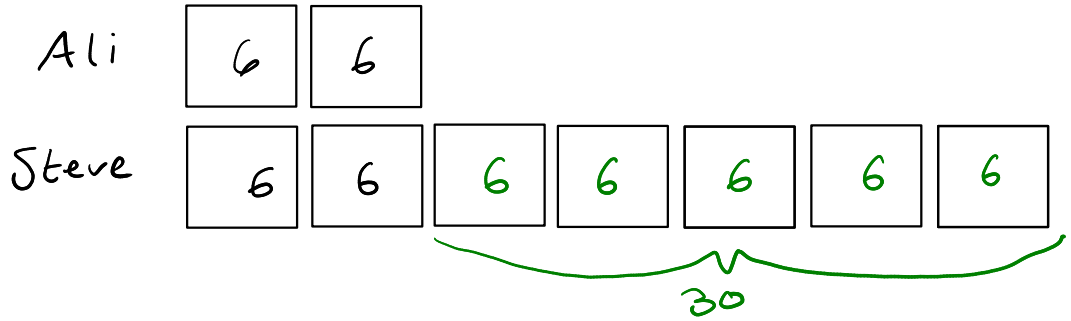
£ 600

(Total for Question 5 is 3 marks)

---

6 Ali and Steve share some sweets in the ratio 2 : 7
Steve gets 30 more sweets than Ali.

Work out how many sweets Steve gets.



30 ÷ 5 = 6

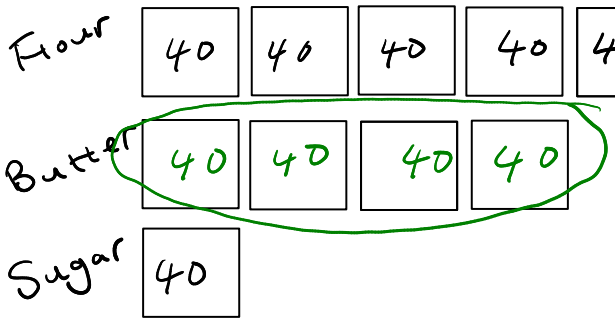
7 x 6 = 42

42

(Total for Question 6 is 3 marks)

7 Dave is making cookies.
He mixes flour, butter and sugar in the ratio 6 : 4 : 1

Dave uses 160 grams of butter.
Work out how much flour and sugar Dave uses.



$6 \times 40 = 240$

$\frac{160}{4} = 40$

flour 240 grams

sugar 40 grams

(Total for Question 7 is 3 marks)

---

8 Alvin and Simon shared £540 in the ratio 4 : 5

Alvin gave half of his share to Theo.
Simon gave a tenth of his share to Theo.

What fraction of the £540 did Theo receive?

$\frac{540}{9} = 60$

A : S

$4 \times 60 : 5 \times 60$

$240 : 300$

$\frac{1}{2} \times 240 = 120$

$\frac{1}{10} \times 300 = 30$

Theo gets $120 + 30 = 150$

$\frac{150}{540} = \frac{15}{54}$

$= \frac{5}{18}$

$\frac{5}{18}$

(Total for Question 8 is 3 marks)

---

9 *ABC* is a straight line.



The length of *BC* is four times the length of *AB*.

BC = 100 metres.
Work out the length *AC*.

AB : BC

1 : 4

25 : 100

$100 \div 4 = 25$

125 metres

(Total for Question 9 is 3 marks)

10 Bob is going to make some orange paint.
He needs to mix red paint, yellow paint and white paint in the ratio 5 : 4 : 1

Bob wants to make 750 m*l* of orange paint.

Bob has

400 m*l* of red paint
300 m*l* of yellow paint
200 m*l* of white paint

$\frac{750}{10} = 75$

Does Bob have enough red paint, yellow paint and white paint to make the orange paint?
You must show all your working.

75 75 75 75 75 : 75 75 75 75 : 75

375 : 300 : 75

375 < 400 ✓ 300 = 300 ✓ 75 < 200 ✓

(Total for Question 10 is 4 marks)

---

11 Megan is going to make a drink using the instructions below.

Mix 2 parts of fruit juice
with 5 parts of sparkling water

Megan has 180 m*l* of fruit juice and 400 m*l* of sparkling water.

What is the greatest amount of the drink Megan can make?

with all fruit juice

90 90 : 90 90 90 90 90

$\frac{180}{2} = 90$ 180 : 450 (not enough water)

with all the water

80 80 : 80 80 80 80 80

$\frac{400}{5} = 80$

160 : 400 160+400

560 ml

(Total for Question 11 is 3 marks)

---

12 In a bag there are only red counters, blue counters and white counters.
A counter is taken at random from the bag.

The table shows the probability of getting a red counter.

Colour	Red	Blue	White
Probability	0.35	0.26	0.39

the number of blue counters : the number of white counters = 2 : 3

Complete the table.

$1 - 0.35 = 0.65$

$0.65 \div 5 = 0.13$ (each part)

Blue : white

$2 \times 0.13 : 3 \times 0.13$

$0.26 : 0.39$

(Total for Question 12 is 4 marks)

---

13 Al, Tom and Joe share €3000.

The ratio of the amount Al gets to the amount Tom gets is in the ratio 5 : 4
Joe gets 1.5 times the amount Tom gets.

$4 \times 1.5 = 6$

Work out the amount of money that Tom gets.

Al : Tom : Joe

5 : 4 : 6

$5 + 4 + 6 = 15$ (15 parts)

$\frac{3000}{15} = 200$

1000 : 800 : 1200

€ 800

(Total for Question 13 is 4 marks)

14 Harry and Gary have a total of 300 stickers.
The ratio of the number of stickers Harry has to the ratio of the number of stickers Gary has is in the ratio 7 : 3 10 parts

Harry gives Gary some stickers.
The ratio of the number of stickers Harry has to the ratio of the number of stickers Gary has is now in the ratio 8 : 7 15 parts

Work out how many stickers Harry gives to Gary.
You must show all your working.

$\frac{300}{10} = 30$

$7 \times 30 : 3 \times 30$

$210 : 90$

$\frac{300}{15} = 20$

$8 \times 20 : 7 \times 20$

$160 : 140$

$210 - 160 = 50$

50

(Total for Question 14 is 4 marks)

15 A shop sells small chocolate bars and large chocolate bars.

There are
small chocolate bars are sold in packs of 4
large chocolate bars are sold in packs of 9

On one day

the number of packs of small chocolate bars sold : the number of packs of large chocolate bars sold = 5 : 2
(×4) (×9)

A total of 190 chocolate bars were sold.

Work out the number of small chocolate bars sold.

Ratio of bars sold :

Small : Large
5 × 4 : 2 × 9
20 : 18 (38 parts)

$\frac{190}{38} = 5$

20 × 5 : 18 × 5
100 : 90

100

(Total for Question 15 is 4 marks)

16 Dermot has 240 counters.
The counters are either red, or blue, or yellow or green.

15% of the counters are red.

$\frac{2}{5}$ of the counters are blue

The ratio of yellow counters to green counters is 3:1

Work out the number of yellow counters Dermot has.

15% × 240 = 36

$\frac{2}{5}$ × 240 = 96

36 + 96 = 132

240 − 132 = 108

Yellow : Green

27	27	27	:	27

$\frac{108}{4} = 27$

27 × 3 = 81

81

(Total for Question 16 is 4 marks)

---

17 Daisy bakes 180 cakes.
She bakes only chocolate cakes, vanilla cakes, banana cakes and lemon cakes.

20% of the cakes are chocolate cakes.

$\frac{4}{9}$ of the cakes are vanilla cakes.

The ratio of banana cakes to lemon cakes is is 3:5

Work out the number of lemon cakes Daisy has.

20% × 180 = 36

$\frac{4}{9}$ × 180 = 80

36 + 80 = 116

180 − 116 = 64

Banana : Lemon

3 : 5

64 ÷ 8 = 8

3 × 8 : 5 × 8

24 : 40

40

(Total for Question 17 is 4 marks)

18 Each month Angelina earns £1800 after tax.

Angelina saves 30% of her salary and she spends the rest.

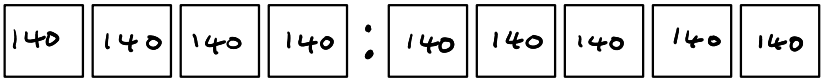
spends 70%

the amount Angelina spends on rent : the amount she spends on other expenses = 4 : 5

Work out how much Angelina spends on rent.

70% × 1800 = 1260

Rent : other



$\frac{1260}{9} = 140$

4 × 140 = 560

£ 560

(Total for Question 18 is 3 marks)

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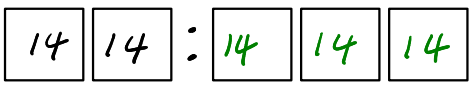
19 There are only red counters and green counters in a bag.

number of red counters : number of green counters = 2 : 3

There are 28 red counters in the bag.

Work out the total number of counters in the bag.

Red : Green



$\frac{28}{2} = 14$

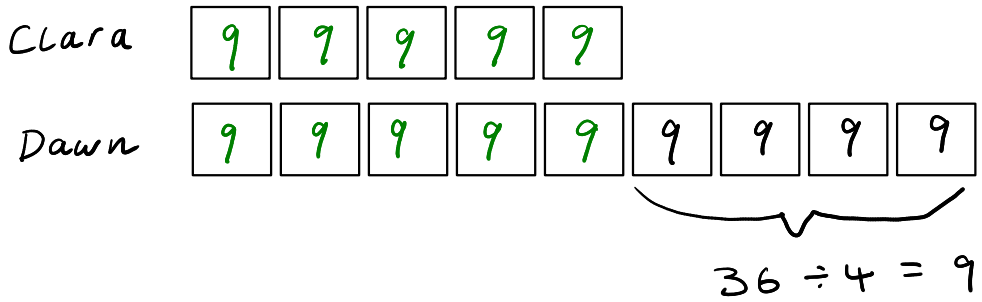
14 × 5 = 70

70

(Total for Question 19 is 2 marks)

20 Clara and Dawn have some stamps in the ratio 5 : 9
Dawn has 36 more stamps than Clara.

Work out the number of stamps Clara has.



5 × 9

45

(Total for Question 20 is 3 marks)

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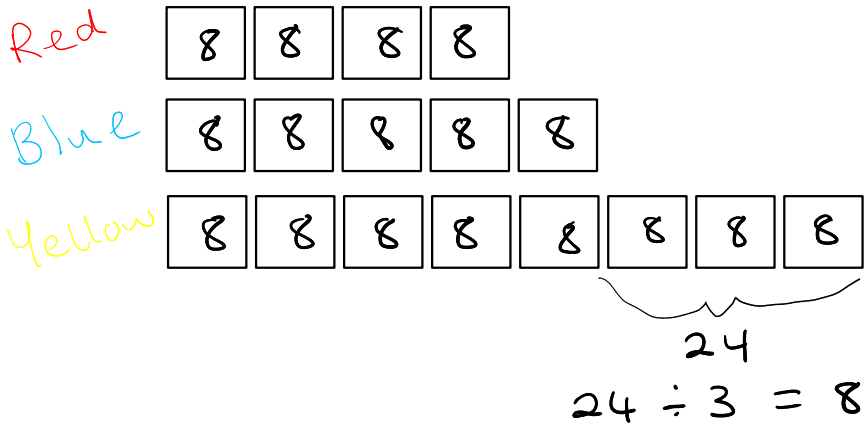
21 In a bag there are counters.
The counters are all either red or blue or yellow.

The number of red counters	:	The number of blue counters	:	The number of yellow counters	=	4 : 5 : 8

17 parts

The number of yellow counters is 24 more than the numbers of blue counters.

Work out the total number of counters in the bag.



17 × 8
= 136

136

(Total for Question 21 is 3 marks)

22 Cameron has some coins with a total value of 95 pence.

She has only 2 pence coins and 5 pence coins.
The ratio

number of 2 pence coins : number of 5 pence coins = 2 : 3

Work out how many 5 pence coins Cameron has.

value of 2p : value of 5p

2 x 2 : 3 x 5

4 : 15

$\frac{95}{19} = 5$

4 x 5 : 15 x 5

20 : 75

20p in 2ps 20 ÷ 2 = 10

75p in 5ps 75 ÷ 5 = 15

15

(Total for Question 22 is 4 marks)

23 Pens and pencils are sold in a shop.

8 pencils cost £1.92
The ratio of the cost of a pen to the cost of a pencil is 4 : 3

Work out the cost of 5 pens.

$\frac{192}{8} = 24$ p for each pencil

24 ÷ 3 = 8

Pen | 8 | 8 | 8 | 8 |

Pencil | 8 | 8 | 8 |

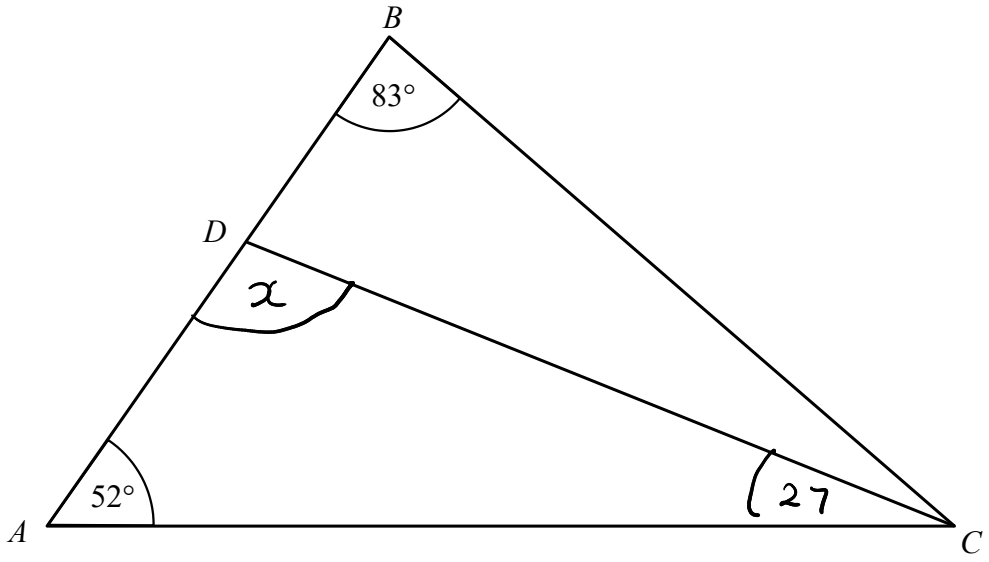
4 x 8 = 32p

0.32 x 5 = £1.60

£ 1.60

(Total for Question 23 is 4 marks)

24 The diagram shows triangle *ABC*.



ADB is a straight line.

the size of angle *BCD* : the size of angle ACD = 2 : 3

Work out the size of angle *ADC*.

BCA = 180 − 52 − 83

= 45°

BCD 9 9

ACD 9 9 9

45 ÷ 5 = 9

BCD : ACD

18 : 27

180 − 27 − 52

= 101°

101 °

(Total for Question 24 is 4 marks)